

Binoy Saha

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EDUCATION

University of Texas at Arlington, Honors College

Data Science

- **GPA:** 3.8
- **Coursework:** Calculus 1 & 2, Intermediate Programming, Python for Data Science 1 & 2, Data Mining

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, HTML/CSS, SQL, Java, R
- **ML/DI Frameworks:** PyTorch, Scikit-learn, TensorFlow, Keras, Hugging Face, OpenAI Whisper, OpenCV, YOLO, RD-DETR, PySR, SciPy, SymPy, Pytorch autograd, Physics Informed Neural Network
- **Data Engineering:** Apache Spark, Apache Kafka, Pandas, NumPy, ETL pipeline, COCO/YOLO data formats
- **MLOps & Backend:** FastAPI, PostgreSQL, Docker, Git, Restful API, JWT

EXPERIENCE

State Farm, ACM

Feb 2026

Undergraduate Researcher

- Built a complete **COCO-to-YOLO** data engineering pipeline in Python processing **10,000+ annotated multi-class** roof damage images - including JSON annotation parsing, bounding box normalization, class remapping, and train/val/test stratified splitting - reducing dataset preparation time by **60%**.
- Implemented **YOLOv8** and **RF-DETR** (transformer-attention) object detection training pipelines with custom config files, benchmarking **precision/recall/mAP** across architectures to quantify the global context advantage of deformable attention over fixed grid cells for subtle damage detection
- Co-authoring research paper comparing CNN vs transformer architectures for UAV hail damage detection, targeting **ACM Multimedia Conference 2026**

ACM @ UT Arlington

Feb 2026

AI AutoReel - ML Backend Developer

- Engineered an end-to-end ML inference pipeline in **FastAPI/PostgreSQL** integrating **Whisper** speech transcription, **librosa** RMS/dB acoustic spike detection, and linguistic keyword scoring to automatically identify sports highlights from raw video, deployed to **10,000+** users
- Built **audio extractor** using **FFmpeg** subprocess orchestration for audio preprocessing, and **spike detection** using **NumPy RMS energy analysis** with configurable dB thresholds and spike merging logic, feeding downstream highlight ranking
- Designed a **keyword fusion scoring algorithm** combining transcript **keyword scores**, **acoustic spike count boosts**, and **intensity-weighted dB** contributions into a unified highlight ranking model, improving highlight precision by removing post-match summary false positives via **regex** pattern filtering
- Implemented JWT authentication backend with bcrypt password hashing and **PostgreSQL** user management, securing all ML inference endpoints behind token-protected middleware

Robomaster Robotics, UTA

Nov 2025

Computer Vision Developer

- Building an end-to-end **YOLOv8** armor plate detection pipeline as only underclassman on team: **extracting and preprocessing** raw video frames from competition footage, **engineering a labeled dataset** via bounding box annotation, and implementing **COCO-to-YOLO data pipeline for model training**
- Architected classical CV baseline using **OpenCV** - HSV masking, contour detection, aspect ratio filtering, confidence-based target prioritization - achieving 30+ FPS on live feeds; **benchmarking against trained YOLO model** to quantify ML vs classical CV tradeoff for real-time embedded deployment at **ARC Championship 2026**

RESEARCH PROJECTS

High Energy Physics Simulation - Quark/Gluon Jet Classification | [GitHub](#)

PyTorch, NumPy, scikit-learn, Pandas, Matplotlib

- Built a complete end-to-end ML research pipeline on **200,000 simulated particle collision jets**: raw data ingestion → reference frame normalization to remove velocity dependence → jet feature engineering (mass, width, energy spread, particle count) → **PyTorch MLP** classifier achieving **79% test accuracy and AUC 0.83**
- Performed permutation feature importance analysis identifying particle count and jet width as top discriminating features - consistent with known physics that gluon jets are broader and more particle-dense than quark jets
- Built as part of Google Summer of Code submission for simulation bias analysis project @ CERN

Physics Informed Neural Network - Damped Harmonic Oscillator

PyTorch, NumPy, Matplotlib

- Implemented a **Physics-Informed Neural Network** that learns to solve a damped spring-mass system across a range of damping conditions - trained using only the governing physics equation as the loss function with zero labeled data, achieving 3–5% error against analytical solutions
- Leveraged **PyTorch autograd** to compute exact derivatives of network outputs with respect to inputs, encoding physical laws directly into training; identified and analyzed spectral bias as a failure mode for high-frequency oscillation cases

Stock Price Prediction - Monte Carlo & ML Pipeline

Python, NumPy, Pandas, scikit-learn, yfinance

- Built an end-to-end data pipeline: **REST API data ingestion via yfinance**, **NumPy/Pandas feature engineering** with 5-day sliding window, **Linear Regression** model achieving **R²=0.95** and **RMSE=\$3.30** on AAPL test data
- Implemented **Monte Carlo simulation with 10,000 stochastic scenarios** to generate confidence intervals, demonstrating probabilistic uncertainty quantification in a production-style **Python** pipeline